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Focal–General Diffusion Model with Semantic Consistent Guidance for Sign Language Production

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Abstract

Sign Language Production (SLP) aims to translate spoken language into sign sequences, where the main challenge lies in generating coherent and natural poses from discrete glosses (G2P). Existing G2P methods typically treat each pose as an indivisible unit, limiting their ability to capture fine-grained joint-level dependencies and thus degrading pose quality. To address this, we propose the Focal–General Diffusion Model (FGDM), characterized by a pioneering two-stage denoising framework that harmonizes local joint-level dependencies and global coherence. Specifically, in the Focal stage, a novel Adaptive Sign GCN (ASGCN) adaptively models each pose based on contextual correlations, skeletal topology, and semantic conditions, ensuring precise generation of local details. In the General stage, a Transformer-based module refines the entire pose sequence to enhance global coherence and naturalness. Moreover, we introduce a Semantic Consistent Guidance (SCG) mechanism that seamlessly integrates semantic supervision into diffusion training, enforcing tighter alignment between generated pose sequences and their intended gloss semantics. Extensive experiments on PHOENIX14T and USTC-CSL demonstrate that FGDM achieves SOTA performance. Our project page is available at here.

1. Introduction

Sign Language Production (SLP) plays a crucial role in promoting communication, education, and the broader adoption of sign languages [2, 3, 46]. However, due to the unique grammatical structures of sign languages and the complex, fine-grained movements they involve, existing methods struggle to generate accurate and realistic sign sequences directly. To address this challenge, previous studies [20, 31, 39, 41] typically decompose the task into two stages for pose generation (shown in Fig. 1 Top): Text-to-Gloss (T2G) and Gloss-to-Pose (G2P). The former has been effectively tackled using NMT-based methods [24, 25], while the latter remains considerably more challenging and thus carries greater research significance. Furthermore, the generated pose sequences from the G2P stage can be used to animate digital avatars or serve as the foundation for sign video synthesis. Motivated by these factors, this work focuses on advancing the G2P task.

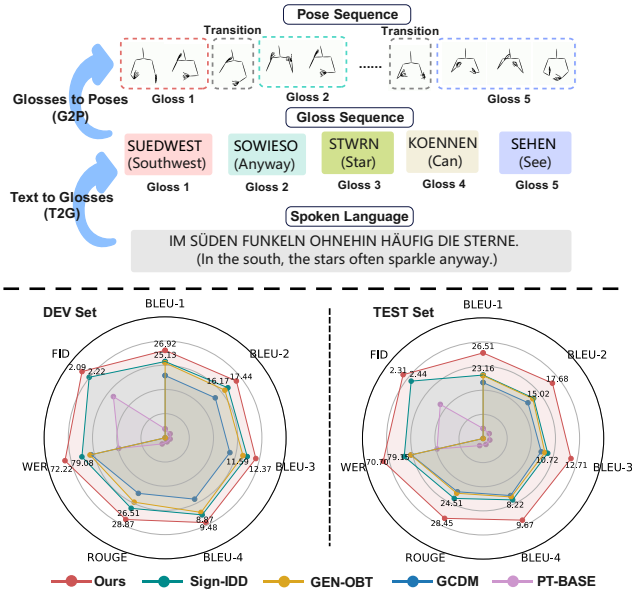


Figure 1. **Top**: Illustration pipeline of SLP, including T2G and G2P. T2G converts text into glosses, while G2P maps each gloss to its corresponding pose sequence and generates smooth transitions to ensure naturalness. This work primarily focuses on G2P. **Bottom**: Comparison with existing methods on the challenging PHOENIX14T [5] dataset. Our approach achieves new SOTA results across all metrics on both DEV and TEST sets.

Recently, increasing efforts have been devoted to improving G2P methods. Saunders *et al.* [31] introduced the first end-to-end approach, a Progressive Transformer for

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G2P, and later enhanced it with adversarial training to produce more realistic results [32, 33]. However, the autoregressive design of Saunders' model leads to error accumulation. To address this issue, Hwang *et al.* [21] proposed a non-autoregressive Transformer that maps gloss sequences directly to pose sequences, while Huang *et al*. [20] further introduced an External Aligner to improve sequence alignment learning in non-autoregressive models. More recently, diffusion-based methods have emerged: Xie *et al.* [47] proposed a discrete diffusion model for G2P, Tang *et al.* [42] enhanced diffusion generation stability through multi-hypothesis aggregation, and in their subsequent work [41], incorporated feature disentanglement into the diffusion process to better capture decoupled attributes. However, existing SOTA methods [39–42, 47] overemphasize global modeling and fail to capture fine-grained joint-level dependencies, leading to degraded pose quality.

Additionally, ensuring the semantic consistency of generated pose sequences is crucial, as it directly determines the intelligibility and expressiveness of sign language generation. However, advanced diffusion architectures [15, 40–42, 47] can only blindly fit coordinates under regression-based supervision, failing to guarantee semantic consistency. Although a few early studies have investigated semantic-guided generation in non-diffusion models [20, 39], these approaches suffer from large cross-modal alignment gaps, which severely affect training stability and overall performance. Moreover, due to the unique characteristics of diffusion training, previous methods cannot be directly transferred to diffusion-based models. In general, how to effectively leverage semantic signals for cross-modal guidance, particularly within diffusion frameworks, remains an open question.

To address the aforementioned challenges, we propose a **Focal–General Diffusion Model** (FGDM) with a **Semantic Consistent Guidance** (SCG) mechanism for the G2P task in SLP. 1) To enhance the model's awareness of fine-grained dependencies, **FGDM** originally adopts a two-stage denoising paradigm: a Focal stage that models joint-level dependencies, and a General stage that enables global sequence coherence. In the Focal stage, we reformulate the graph convolution and introduce the Adaptive Sign GCN (ASGCN), which innovatively incorporates contextual correlations and semantic conditions for adaptive pose modeling. In the General stage, skeletal representations are projected into a higher-level space, allowing the Transformer-based module to enhance global naturalness and coherence. 2) To bridge the gap in cross-modal semantic supervision, SCG introduces a new perspective: it jointly trains a semantic decoder that maps the generated pose features into the semantic space. In this space, pose features are effectively aligned with gloss sequences by maximizing the likelihood over all feasible alignment paths, following the principle of Connectionist Temporal Classification (CTC) [12]. Moreover, to adapt to diffusion training, SCG dynamically adjusts the supervision strength according to the current sampling stage, ensuring stable model convergence. As shown in Fig. 1, FGDM trained with SCG achieves **SOTA** results across all metrics on the challenging PHOENIX14T benchmark, confirming the effectiveness of our method.

2. Related Work

2.1. Sign Language Production

Sign language production (SLP), as the inverse of recognition [14, 18, 19, 29, 51] and translation [7, 8, 53], directly benefits the Deaf community and also supports the promotion and teaching of sign language. With the increasing demand, research interest in SLP has been steadily growing [30, 45].

Early approaches to SLP relied on avatar-based technologies [11, 23] and Statistical Machine Translation (SMT) [24, 25], but these methods were costly and inflexible. Stoll *et al*. [38] first introduced deep learning and decomposing SLP into three subproblems: Text-to-Gloss (T2G), Gloss-to-Pose (G2P), and Pose-to-Video (P2V), with G2P receiving the most attention due to its cross-modal challenges and practical value. Saunders *et al.* [31] proposed the first end-to-end G2P model, Progress Transformer, and later enhanced it with GANs and Mixture Density Networks [32, 34]. Subsequent research [20, 21] has underscored the inherent limitations of autoregressive models (e.g., error accumulation, regression to the mean), thereby motivating the exploration of non-regressive alternatives. Recently, diffusion models have been applied to G2P. Xie *et al.* [47] employed discrete denoising diffusion for discretized pose representations, while Tang *et al.* [41] introduced an iconicity-disentangled diffusion framework that maps 3D joints into 4D bones. However, current methods typically ignore joint-level dependencies and solely rely on regression losses, which severely degrade generation quality. To overcome these limitations, we introduce FGDM with an SCG supervision mechanism.

2.2. Graph Convolutional Networks

Graph Convolutional Networks (GCNs) [4] can effectively capture both topological structures and node attributes in graph data, making them well-suited for skeleton-based tasks. In Skeleton-Based Action Recognition, Yan *et al.* [48] introduced ST-GCN, the first framework to leverage graph convolution for modeling spatiotemporal skeleton dynamics. This pioneering work inspired a series of follow-up studies [9, 10, 26, 36, 37]. Among them, Shi *et al.* [36] introduced the Adaptive Graph Convolutional Network (AGCN), where the adjacency matrix is dynamically learned rather than predefined. It was later extended with a multi-stream architecture and refined attention mechanisms

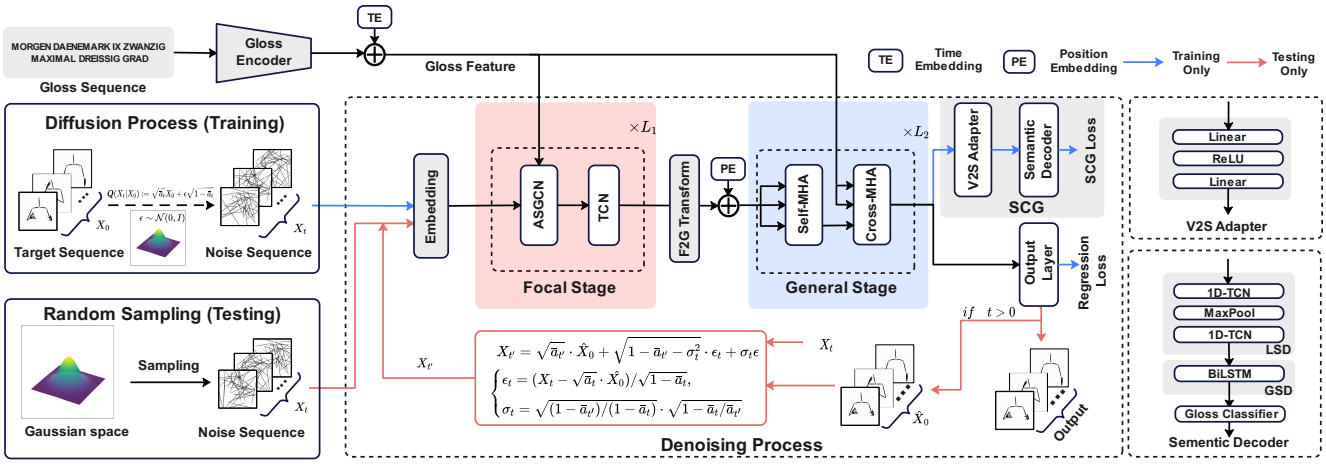


Figure 2. Overall pipeline of FGDM during training and inference. In training, the noisy sequence X_t is generated by adding noise to the target sequence X_0 for t steps. FGDM denoises X_t under glosses guidance to predict the initial sequence $\hat{X}_0$. It comprises two stages: a Focal stage for joint-level dependency modeling and a General stage for global sequence modeling. Beyond the standard regression loss, Semantic Consistent Guidance (SCG) provides auxiliary supervision, where the V2S Adapter and Semantic Decoder project visual features into the semantic space, and the SCG loss enforces tighter consistency with the gloss sequence. During inference, FGDM iteratively denoises random inputs to generate sign sequences.

to further boost performance [37]. In sign language recognition (SLR), ST-GCN has been widely adopted. Jiao *et al.* [22] leveraged multiple ST-GCNs to capture region-specific body dynamics, whereas Li *et al.* [27] adopted ST-GCN as a robust pose encoder. For SLP, Saunders *et al.* [35] developed a graph-based pose decoder, Huang *et al.* [20] integrated ST-GCN as the generator head, and He *et al.* [15] proposed a diffusion-based architecture built upon ST-GCN. However, the frame-shared GCN paradigm is suboptimal for SLP, as joint semantics and spatial dependency strength dynamically change over time, making static methods incapable of adapting flexibly. To address this, we propose a novel frame-wise adaptive GCN, ASGCN, to achieve flexible and precise sign pose modeling in the Focal stage.

3. Methodology

3.1. Overview

Given a gloss sequence $\mathcal{G} = \{g^i\}_{i=1}^{L}$ containing L glosses, our objective is to generate the corresponding sign pose sequence $\mathcal{X} = \{x^i\}_{i=1}^{S}$ with S frames, where each $x^i \in \mathbb{R}^{J\times 3}$ represents the 3D coordinates of J keypoints. As illustrated in Fig. 2, the proposed **Focal–General Diffusion Model (FGDM)** presents the overall architecture as well as the training and inference pipelines. The key innovation of this framework lies in constructing a novel denoising network, enabling progressive modeling from local joint dependencies to global sequence coherence. Moreover, the **Semantic Consistent Guidance (SCG)** ensures consistency between the generated sequence and the semantic conditions. The following describes each component.

3.2. Focal–General Diffusion Model

As a diffusion-based model, FGDM comprises two Markov chains during training: a **diffusion process** and a **denoising process**. In the **diffusion process**, the 3D target pose sequence $X_0 \in \mathbb{R}^{S\times J\times 3}$ is gradually corrupted by Gaussian noise $\epsilon \sim \mathcal{N}(0, \mathbf{I})$ for t steps, resulting in a noisy sequence X_t. Following DDPMs [16], this process is formulated as:

$$Q(X_t|X_0) := \sqrt{\bar{a}_t}X_0 + \epsilon\sqrt{1-\bar{a}_t}, \quad t \sim U(0,T), \tag{1}$$

where T denotes the maximum diffusion sample steps, $\bar{a}_t := \prod_{s=0}^{t} a_s$, and $a_s := 1 - \beta_s$. Here, β_s represents the cosine noise variance schedule. During inference, X_t is directly sampled from a Gaussian distribution. As the main innovation lies in the **denoising process**, we focus on introducing this part as follows.

Embedding. We first apply Iconicity Disentanglement [41] to $X_t \in \mathbb{R}^{S\times J\times 3}$ to obtain an enhanced representation $X'_t \in \mathbb{R}^{S\times J\times 7}$. Then, X'_t is embedded into $X^e_t \in \mathbb{R}^{S\times J\times d}$ through an embedding matrix. Meanwhile, the gloss sequence is processed by the Gloss Encoder [42], whose output is fused with the time embedding to generate the gloss feature $G \in \mathbb{R}^{L\times c}$.

Focal Stage. As illustrated in Fig. 2, the Focal stage primarily comprises L_1 stacked layers of ASGCN and TCN. ASGCN primarily models spatial joint dependencies, while TCN focuses on temporal joint dynamics, following the space–time decomposition design [36, 48]. Since ASGCN is our core contribution, we detail it below.

ASGCN. Conventional static graph convolutions [26, 36, 37, 48] are insufficient to model the dynamic variations in joint semantics and spatial dependencies inherent in sign

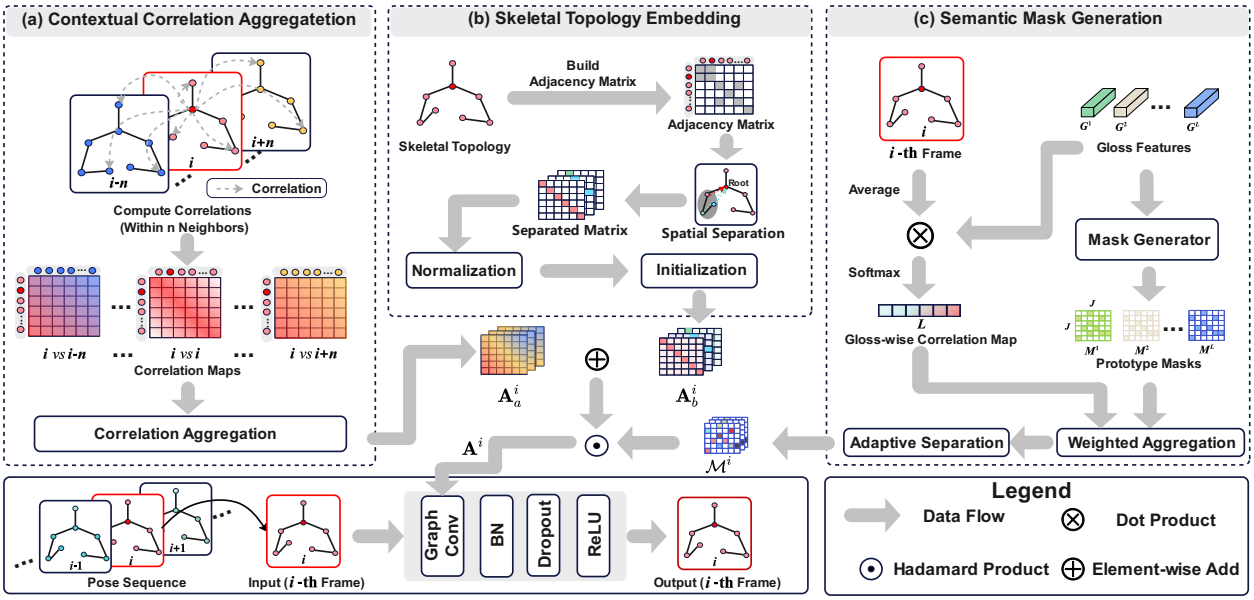


Figure 3. Illustration of ASGCN. Taking the i-th frame as an example, ASGCN constructs a frame-wise adaptive adjacency matrix by integrating (a) contextual correlations A_a^i and (b) skeletal topology A_b^i. The (c) semantic mask $\mathcal{M}^i$ further enhances or suppresses the connection strengths in $A_a^i + A_b^i$ from a semantic perspective. Graph convolution is then applied, followed by batch normalization (BN), dropout, and ReLU activation.

language. To address this, we first design a more flexible graph convolution operation:

$$X_{out}^i = \mathbf{W}_\theta \sum_k^{K_v} \mathbf{W}_{\delta k} \odot (X_{in}^i \mathbf{A}_k^i), \tag{2}$$

where X_{in}^i and X_{out}^i denote the input and output of the i-th frame, respectively; K_v is the kernel size and typically set to 3; $\mathbf{W}_{\delta k} \in \mathbb{R}^{C_{in} \times 1 \times 1 \times 1}$ and $\mathbf{W}_\theta \in \mathbb{R}^{C_{out} \times C_{in} \times 1 \times 1}$ represent the depthwise convolution and the pointwise convolution, respectively, following the separable design [17]. $\mathbf{A}_k^i \in \mathbb{R}^{J \times J}$ denotes the k-th adjacency matrix of the i-th frame, where $\odot$ represents the Hadamard product.

According to the above formulation, since the adjacency matrix $\mathbf{A}^i$ is no longer shared, we design $\mathbf{A}^i$ to be frame-level adaptive and inject semantic conditions as follows:

$$\mathbf{A}^i = (\mathbf{A}_a^i + \mathbf{A}_b^i) \odot \mathcal{M}^i. \tag{3}$$

Here, $\mathbf{A}_a^i$ and $\mathbf{A}_b^i$ are constructed based on contextual correlation and topological dependencies, respectively, while $\mathcal{M}^i$ serves as a dynamic semantic mask that enhances or suppresses joint connections from a semantic perspective. We next describe these three components in detail.

(a) Get $\mathbf{A}_a^i$. Modeling multi-frame contextual correlations enables the GCN to infer dynamic, context-aware joint dependencies, resulting in more coherent and accurate representations for the current frame. As shown in Fig. 3 (a), for the i-th frame X^i, we compute its correlations with itself and its n neighboring frames as follows:

$$\mathbf{C}^i = \{ \mathbf{W}_\psi X^i \cdot \mathbf{W}_\tau X^{i+j} \mid j \in [-n, n] \}, \tag{4}$$

where $\mathbf{C}^i \in \mathbb{R}^{(2n+1) \times J \times J}$, and $\mathbf{W}_\psi$ and $\mathbf{W}_\tau$ denote two distinct convolution for embedding. We then aggregate the $(2n+1)$ correlation matrices in $\mathbf{C}^i$ to obtain $\mathbf{A}_a^i$:

$$\mathbf{A}_a^i = \sigma(\mathbf{C}^i) \cdot \mathbf{W}_{Agg}, \tag{5}$$

where the sigmoid function $\sigma(\cdot)$ serve as a gating mechanism and ensures numerical stability. $\mathbf{W}_{Agg} \in \mathbb{R}^{(2n+1) \times K_v}$ denotes a linear layer that adaptively weights and integrates correlations. Note that $\mathbf{W}_{Agg}$ is initialized to zero, allowing the model to gradually incorporate $\mathbf{A}_a^i$.

(b) Get $\mathbf{A}_b^i$. The purpose of $\mathbf{A}_b^i$ is to preserve the skeletal prior topology and enhance structural stability. As shown in Fig. 3 (b), we first set the neck joint as the root to establish the skeletal hierarchy. We then apply Spatial Separation [48] to decompose the topology into K_v submatrices, which are subsequently normalized to initialize $\mathbf{A}_b^i$. As $\mathbf{A}_b^i$ encodes the physical topology, it remains consistent and is shared across all frames.

(c) Get $\mathcal{M}^i$. Since sign poses with different semantics may exhibit distinct joint dependency strengths, we construct a semantic mask $\mathcal{M}^i$ to align the graph structure with the corresponding semantics. As shown in Fig. 3 (c), for a gloss feature sequence $G = G^1, G^2, \ldots, G^L$, we build a corresponding mask set $M = M^1, M^2, \ldots, M^L$, which serves as prototype masks for generating $\mathcal{M}^i$:

$$M = \{\mathcal{MG}(G^l) \mid l \in [1, L]\}, \tag{6}$$

the mask generator $\mathcal{MG}(\cdot)$ maps each gloss feature into a $J \times J$ mask, and this mapping is currently realized through

a linear layer, with potential improvements left for future exploration. For the i-th frame X^i, we compute the mean of its J joints and then calculate its correlation with each gloss feature, followed by a softmax operation:

$$w^i = Softmax((\mathbf{W}_{\psi'} \frac{1}{J} \sum_{j=1}^{J} X_j^i) \cdot (\mathbf{W}_{\tau'} G)), \tag{7}$$

here $\mathbf{W}_{\psi'}$ and $\mathbf{W}_{\tau'}$ denote two distinct 1×1 convolutions for feature embedding. We then aggregate the prototype matrices using the weights $w^i \in \mathbb{R}^L$ to obtain a weighted representation $\bar{M}^i \in \mathbb{R}^{J \times J}$:

$$\bar{M}^i = \sum_{l=1}^{L} w^{il} M^l. \tag{8}$$

Subsequently, we employ a convolution $\mathbf{W_s}$ to adaptively simulate the Spatial Separation [48], and a scaled sigmoid maps the mask values to [0, 2], enabling suppression (< 1) or enhancement (> 1):

$$\mathcal{M}^i = 2 \cdot \sigma(\mathbf{W_s} \bar{M}^i). \tag{9}$$

After obtaining $\mathbf{A}_a^i$, $\mathbf{A}_b^i$, and $\mathcal{M}^i$, we compute the final adjacency matrix $\mathbf{A}^i$ using Eq. 3, apply graph convolution as defined in Eq. 2, and then perform batch normalization, dropout, and ReLU to produce the output of ASGCN.

F2G Transform. Since the General stage no longer focuses on joint-level dependencies, we transform the Focal output $X_t^{f_o}$ as follows:

$$X_t^g = T_{F \to G}(X_t^{f_o}) + PE, \tag{10}$$

where $T_{F \to G}(\cdot)$ denotes a transformation function that converts the feature shape from $S \times J \times d_f$ to $S \times d_g$, mainly consisting of reshape operations and linear transformations. PE denotes the temporal positional encoding.

General Stage. The General stage models long-range dependencies to promote globally coherent and natural sign sequences. As shown in Fig. 2, it is implemented as a Transformer decoder [43] with L_2 stacked layers, and its main computation is given by:

$$\begin{aligned} X_t^{g'} &= MHA(Q, K, V)\,|_{Q=K=V=X_t^g}, \\ X_t^{g_o} &= MHA(Q, K, V)\,|_{Q=X_t^{g'},\ K=V=G}, \end{aligned} \tag{11}$$

where $MHA(Q, K, V)$ denotes the multi-head attention mechanism [43], which is not elaborated here as it is not the focus of this work.

Output Layer. For the output $X_t^{g_o}$, we apply layer normalization and a Multi-Layer Perception (MLP) to obtain the final prediction:

$$\hat{X}_0 = MLP(LayerNorm(X_t^{g_o})). \tag{12}$$

During inference, $\hat{X}_0$ is used to estimate $X_{t'}$ as follows:

$$X_{t'} = \sqrt{\bar{a}_{t'}} \cdot \hat{X}_0 + \sqrt{1 - \bar{a}_{t'} - \sigma_t^2} \cdot \epsilon_t + \sigma_t \epsilon, \tag{13}$$

where t and t' denote the current and next timesteps, respectively, with the initial $t = T$. $\epsilon \sim \mathcal{N}(0, I)$ is a standard Gaussian noise independent of X_t, and

$$\begin{cases} \epsilon_t = (X_t - \sqrt{\bar{a}_t} \cdot \hat{X}_0)/\sqrt{1 - \bar{a}_t}, \\ \sigma_t = \sqrt{(1 - \bar{a}_{t'})/(1 - \bar{a}_t)} \cdot \sqrt{1 - \bar{a}_t/\bar{a}_{t'}}. \end{cases} \tag{14}$$

Here, ϵ_t denotes the noise at timestep t (see Eq. 1), while σ_t controls the stochasticity of the diffusion process. In the next iteration, $X_{t'}$ is fed into FGDM to progressively refine $\hat{X}_0$ over I steps. Starting from $t = T$, the timestep is updated as $t = T \cdot (1 - i/I)$, where $i \in [0, I)$, and I balances the trade-off between diversity and quality of the results.

3.3. Semantic Consistent Guidance

SCG uniquely introduces semantic guidance into diffusion training, bridging the cross-modal gap and ensuring strong semantic consistency in the generated pose sequences.

As shown in Fig. 2, for the visual features $X_t^{G_o}$ output from the General stage, the V2S Adapter first performs an intermediate visual-to-semantic transition:

$$X_t^{trans} = V2S_Adapter(X_t^{g_o}), \tag{15}$$

where $V2S_Adapter(\cdot)$ consists of two linear layers and a ReLU activation. $X_t^{trans} \in \mathbb{R}^{S \times d_g}$ is then decoded into the semantic space by the Semantic Decoder.

The architecture of the Semantic Decoder is inspired by advances in sign language recognition [19, 28, 49, 50]. As illustrated in Fig. 2, it consists of three main components: a Local Semantic Decoder (LSD), a Global Semantic Decoder (GSD), and a Gloss Classifier, defined as follows:

$$\begin{aligned} y_t^l &= LSD(X_t^{trans}), \\ y_t^g &= GSD(y_t^l), \\ y_t^o &= Gloss_Classifier(y_t^g). \end{aligned} \tag{16}$$

$LSD(\cdot)$ consists of two 1D-TCNs and a MaxPool layer to model local features, producing $y_t^l \in \mathbb{R}^{S' \times d_g'}$ with $S' < S/2$. $GSD(\cdot)$ employs a BiLSTM to capture global dependencies, and the final outputs is $y_t^o \in \mathbb{R}^{S' \times (num_glosses+1)}$, where $num_glosses + 1$ denotes all glosses plus an additional $blank$ label for alignment.

Given the logits y_t^o and the corresponding glosses $\mathcal{G}$, we reformulate the CTC loss [12] to obtain the SCG loss:

$$\mathcal{L}_{\text{SCG}} = -\frac{1}{e^{\alpha \frac{t}{T}}} \log \sum_{\pi \in \mathcal{B}^{-1}(\mathcal{G})} p(\pi \mid y_t^o), \tag{17}$$

where $\frac{1}{e^{\alpha \frac{t}{T}}}$ is used to make the model aware of the current diffusion stage t, controlling the loss magnitude to ensure model stable convergence. $\mathcal{B}^{-1}(\mathcal{G})$ denotes the set of all alignments π of length S' that can be collapsed into $\mathcal{G}$ via the dynamic programming algorithm [12]. The probability of an alignment path π is computed under the assumption of conditional independence across time steps:

$$p(\pi \mid y_t^o) = \prod_{s=1}^{S'} p(\pi_s \mid y_t^o), \tag{18}$$

where the per-frame probabilities $p(\pi_s \mid y_t^o)$ is obtained via a softmax operation over the logits y_t^o.

Table 1. Comparison with existing methods on PHOENIX14T. '†' indicates reproduced results, while '–' denotes results missing in the original paper. Best results are in bold, and the second-best are underlined.

Methods	DEV							TEST						
	B1↑	B2↑	B3↑	B4↑	ROUGE↑	WER↓	FID↓	B1↑	B2↑	B3↑	B4↑	ROUGE↑	WER↓	FID↓
Ground Truth	29.77	20.21	15.16	12.13	29.60	74.17	0.00	29.76	20.12	14.93	11.93	28.98	71.94	0.00
Non-Diffusion Based														
PT-base† [31] $_{ECCV'20}$	10.13	5.43	4.25	3.62	11.20	100.00	3.25	10.90	6.10	4.89	4.25	11.14	100.00	3.62
PT-GN† [31] $_{ECCV'20}$	14.30	7.53	4.58	3.15	12.91	94.00	2.66	15.07	8.73	5.77	4.25	14.08	92.30	2.91
NAT-EA [20] $_{MM'21}$	–	–	–	–	–	–	–	15.12	10.45	7.99	6.66	19.43	82.01	–
GEN-OBT [39] $_{MM'22}$	24.92	15.72	11.20	8.68	25.21	82.36	–	23.08	14.91	10.48	8.01	23.49	81.78	–
DET [44] $_{SEM'23}$	17.25	10.17	7.04	5.32	17.85	–	–	17.18	10.39	7.39	5.76	17.64	–	–
Diffusion Based														
G2P-DDM [47] $_{AAAI'24}$	–	–	–	–	–	–	–	16.11	11.37	9.22	7.50	–	77.26	–
GCDM [42] $_{TOMM'24}$	22.88	14.28	10.01	7.64	23.35	82.81	–	22.03	14.21	10.16	7.91	23.20	81.94	–
Sign-IDD† [41] $_{AAAI'25}$	25.13	16.17	11.59	8.87	26.51	79.08	2.22	23.16	15.02	10.72	8.22	24.51	79.15	2.44
Sign-D2C [40] $_{CVPR'25}$	22.46	–	–	8.06	22.07	80.73	2.68	22.36	–	–	8.73	22.36	79.06	2.98
FGDM (ours)	**26.92**	**17.44**	**12.37**	**9.48**	**28.87**	**72.22**	**2.09**	**26.51**	**17.68**	**12.71**	**9.67**	**28.45**	**70.70**	**2.31**

Table 2. Comparison with existing methods on USTC-CSL. Split-I and Split-II refer to two partitioning schemes [13]. '†' indicates reproduced results. Best results are in bold, and the second-best are underlined.

Methods	TEST (Split-I)							TEST (Split-II)						
	B1↑	B2↑	B3↑	B4↑	ROUGE↑	WER↓	FID↓	B1↑	B2↑	B3↑	B4↑	ROUGE↑	WER↓	FID↓
Ground Truth	96.08	95.51	95.23	95.01	95.90	4.24	0.00	44.01	26.13	13.74	0.00	43.14	68.50	0.00
PT-base† [31] $_{ECCV'20}$	27.29	10.19	4.77	3.27	29.71	98.77	0.28	11.41	0.00	0.00	0.00	15.23	96.45	0.41
PT-GN† [31] $_{ECCV'20}$	30.06	15.60	10.35	8.51	28.37	88.10	0.22	23.00	9.99	0.00	0.00	26.10	78.36	0.43
Sign-IDD† [41] $_{AAAI'25}$	90.53	87.31	85.47	83.88	91.67	14.01	**0.07**	41.86	13.43	2.68	0.00	41.01	74.41	0.26
FGDM (ours)	**92.04**	**89.69**	**88.58**	**87.80**	**92.37**	**12.98**	**0.07**	**43.61**	**26.46**	**15.71**	0.00	**42.19**	**72.00**	**0.25**

Loss Design. The overall loss of FGDM is defined as:

$$\mathcal{L} = \underbrace{\mathcal{L}_{joint} + \lambda \cdot \mathcal{L}_{bone}}_{\text{Regression Loss}} + \gamma \cdot \mathcal{L}_{SCG}, \tag{19}$$

where λ and γ are weighting coefficients. $\mathcal{L}_{SCG}$ denotes the proposed SCG loss introduced in Eq. 17, while $\mathcal{L}_{joint}$ and $\mathcal{L}_{bone}$ are conventional regression losses, defined as:

$$\begin{cases} \mathcal{L}_{joint} = \frac{1}{S}\sum_{s=1}^{S} \|X_0^s - \hat{X}_0^s\|_1, \\ \mathcal{L}_{bone} = \frac{1}{S}\sum_{s=1}^{S} \|Bone(X_0^s) - Bone(\hat{X}_0^s)\|_2^2. \end{cases} \tag{20}$$

Here, X_0^s and $\hat{X}_0^s$ denote the ground-truth and predicted poses at the s-th frame, respectively, and $Bone(\cdot)$ extracts bone vectors based on the skeletal topology.

4. Experiments

4.1. Experimental Setup

4.1.1. Datasets

PHOENIX14T [5] is the most widely used dataset for SLP. It comprises 2,887 unique German words and 1,066 distinct signs, totaling 8,257 German weather forecast samples with corresponding annotations. The dataset is partitioned into 7,096 training, 519 validation, and 642 testing samples.

USTC-CSL [13] is a Chinese sign language dataset that has been adopted in recent SLP studies [15, 40, 41]. It contains 100 unique sentences performed by 50 signers and provides two official data splits: Split-I (4,000 training and 1,000 testing samples) and Split-II (4,700 training and 300 testing samples). Previous SLP methods [15, 40, 41] used only Split-I, where test sentences overlap with the training set. In this work, we evaluate on both Split-I and the more challenging Split-II.

4.1.2. Evaluation Metrics.

Following previous works [20, 39, 42, 47], the generated sign poses are back-translated into gloss sequences and text using a sign language translation model [5]. The outputs are then evaluated against the references using WER, BLEU, and ROUGE. Moreover, FID is also employed to assess the quality of the generated results. These evaluation metrics are described in more detail in the supplementary materials.

4.1.3. Implementation Details.

Following prior works [20, 39, 42, 47], we first use OpenPose [6] to extract 50 human 2D keypoints per frame from PHOENIX14T. The keypoints are then converted into 3D coordinates using a skeletal correction model [52] to serve as the ground truth. For USTC-CSL, which provides 25 human 3D keypoints per frame, normalization is applied, and the processed data are used as the ground truth. For model evaluation, we adopt NSLT [5], which translates the generated pose sequences into gloss sequences and spoken language. To ensure fair comparison, we use the NSLT weights from [41] on PHOENIX14T. Since pretrained weights for USTC-CSL are unavailable, we retrain the model and reproduce all compared methods for consistent and fair evaluation. In addition, the diffusion model is trained with 1000

Table 3. Ablations for the main innovations on PHOENIX14T, where the General stage is regarded as our baseline.

Methods	DEV			TEST		
	B1↑	B4↑	WER↓	B1↑	B4↑	WER↓
Baseline (General only)	22.54	7.39	81.00	22.41	7.77	79.76
Baseline+Focal	**28.09**	**9.84**	76.57	26.29	9.25	73.91
Baseline+SCG	25.13	9.12	78.92	24.26	9.17	78.12
Baseline+Focal+SCG	26.92	9.48	**72.22**	**26.51**	**9.67**	**70.70**

Table 4. Ablations for parameters L_1 and L_2 on PHOENIX14T.

Methods		DEV			TEST		
		B1↑	B4↑	WER↓	B1↑	B4↑	WER↓
L_1	1	26.10	9.17	76.27	24.92	9.13	75.72
	2	26.53	8.93	72.40	25.33	9.16	71.45
	3	**26.92**	**9.48**	**72.22**	26.51	**9.67**	70.70
	4	26.46	9.21	72.35	**26.53**	9.64	**70.42**
L_2	1	24.12	7.54	79.13	23.43	8.27	77.44
	2	**26.92**	**9.48**	**72.22**	**26.51**	**9.67**	**70.70**
	3	23.72	8.24	73.79	23.29	8.51	72.62

timesteps ($T = 1000$) and 5 inference steps ($I = 5$). We use the Adam optimizer [1] with a learning rate of 0.001. The n in Eq. 4 is set to 3, α in Eq. 17 is set to 10, while λ and γ in Eq. 19 are set to 0.1 and 0.0001, respectively. All experiments are conducted on 1 NVIDIA A6000 GPU.

4.2. Comparison with State-of-the-Arts

We evaluate FGDM on two datasets, PHOENIX14T and USTC-CSL.

PHOENIX14T. Tab. 1 presents the comparison between FGDM and existing SLP methods on the PHOENIX14T. According to whether diffusion models are used, methods are grouped into *Non-Diffusion Based* and *Diffusion Based*. FGDM consistently outperforms all baselines across all metrics, achieving new SOTA results. Specifically, compared with the strongest non-diffusion method GEN-OBT, FGDM shows notable gains of +2.0%/+3.43% (DEV/TEST) on BLEU-1, +3.66%/+4.96% on ROUGE, and -10.14%/-11.08% on WER. Even among diffusion-based models, FGDM remains dominant, surpassing the best-performing Sign-IDD by +2.36%/+3.94% in ROUGE and reducing WER by -6.86%/-8.45%. Interestingly, FGDM achieves a lower WER than Ground Truth, likely because 1) its generated poses are closer to the training distribution, aiding recognition, and 2) it can correct mislabeled ground truth samples (see Fig. 4 (c)). These comparison results highlight the strong performance of FGDM.

USTC-CSL. Since most existing methods only report results on PHOENIX14T, we compare FGDM with available open-source methods on USTC-CSL. As shown in the Tab. 2, FGDM consistently outperforms others on both Split-I and Split-II. Specifically, on Split-I, although FGDM and Sign-IDD achieve the same FID of 0.07, FGDM surpasses Sign-IDD on all other metrics, with BLEU-1 and BLEU-4 improved by +1.51% and +3.92%, respectively, and WER reduced by -1.03%. On the more challenging Split-II, while all methods (including the ground truth) yield a BLEU-4 of 0, FGDM still leads in the remaining metrics, notably outperforming Sign-IDD by +13.01% and +13.03% on BLEU-2 and BLEU-3. These results demonstrate that FGDM achieves strong performance and generalization across different languages and dataset splits.

Table 5. Ablations for $\mathbf{A}_a^i$, $\mathbf{A}_b^i$, $\mathcal{M}^i$ in Eq. 3 on PHOENIX14T.

Methods	DEV			TEST		
	B1↑	B4↑	WER↓	B1↑	B4↑	WER↓
Baseline+Focal	**28.09**	**9.84**	**76.57**	**26.29**	**9.25**	**73.91**
w/o $\mathbf{A}_a^i$	23.29	7.99	79.16	22.13	7.88	78.47
w/o $\mathbf{A}_b^i$	26.44	9.08	76.70	25.05	8.80	75.18
w/o $\mathcal{M}^i$	25.00	8.69	77.66	23.46	8.40	77.20

4.3. Ablation Study

We perform ablations on the PHOENIX14T to evaluate the proposed methods.

Ablations for the main innovations. As shown in Tab. 3, we take the General stage as the baseline and progressively incorporate the other components. According to the results, both components consistently enhance performance. Introducing the Focal stage boosts BLEU-1 to 28.09% and BLEU-4 to 9.84% on the DEV, while reducing WER by -4.43%/-5.85% (DEV/TEST). Adding the SCG strategy also significantly improves all metrics. When Focal and SCG are combined, the model achieves the best results, with WER reductions of -8.78%/-9.06% compared to the baseline. Overall, both Focal and SCG contribute to higher-quality pose generation, and their combination yields the best results.

Ablations for parameters L_1 and L_2. We investigate the impact of the stacked layer numbers in the Focal and General stages. As shown in Tab. 4, increasing L_1 consistently improves performance, with $L_1 = 3$ achieving the best BLEU-1, BLEU-4, and WER on DEV and optimal BLEU-4 on TEST. Further increases bring negligible gains (+0.02% BLEU-1, –0.28% WER). For L_2, the best results are obtained when $L_2 = 2$. Therefore, we adopt $L_1 = 3$ and $L_2 = 2$ as the final configuration.

Ablations for $\mathbf{A}_a^i$, $\mathbf{A}_b^i$, and $\mathcal{M}^i$ in Eq. 3. As shown in Tab. 5, we ablate the core components of Eq. 3 in ASGCN. $\mathbf{A}_a^i$, $\mathbf{A}_b^i$, and $\mathcal{M}^i$ are generated based on contextual correlations, skeletal topology, and semantic conditions, respectively. Removing any of them degrades performance. Excluding $\mathbf{A}_a^i$ causes the largest drop (BLEU-1: -4.80%/-4.16%; WER: +2.59%/+4.56%), followed by $\mathcal{M}^i$ (BLEU-1: -3.09%/-2.83%; WER: +1.09%/+3.29%). Above results demonstrate the effectiveness of combining $\mathbf{A}_a^i$, $\mathbf{A}_b^i$, and $\mathcal{M}^i$, with the correlation term $\mathbf{A}_a^i$ and semantic mask $\mathcal{M}^i$ being most critical to ASGCN's performance.

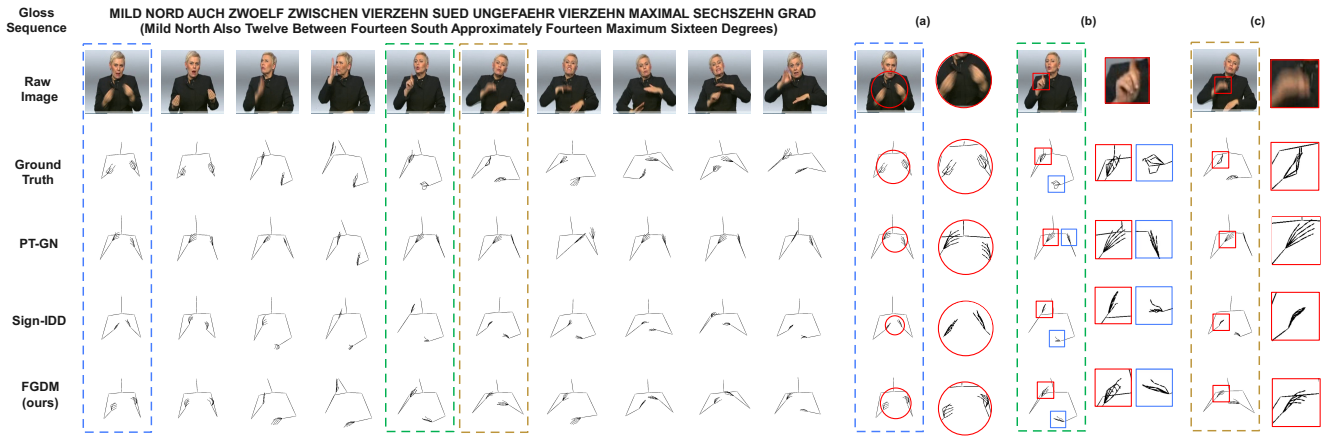


Figure 4. Qualitative comparison of generation quality with previous methods on PHOENIX14T. We compare FGDM with PT [31] and Sign-IDD [41], together with the corresponding gloss sequence, raw images, and ground truth (GT) poses. Panels (a)–(c) present zoomed-in regions where FGDM generates more natural and GT-consistent hand details. Even when the GT is inaccurate (as in (b) (c)), FGDM still produces realistic results, demonstrating its superior performance.

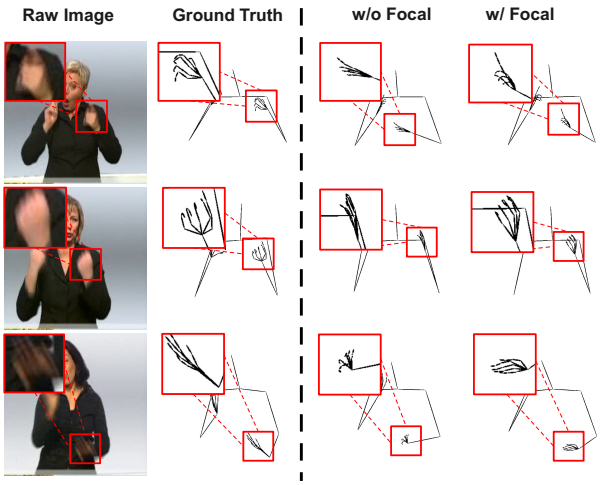


Figure 5. Qualitative comparison of generation quality without and with the Focal stage (w/o Focal vs. w/ Focal). Hand details are enlarged for better observation.

4.4. Qualitative Results

Qualitative comparison with previous methods. As shown in Fig. 4, we compare the generated results of FGDM with PT-GN and Sign-IDD on the PHOENIX14T. Since PT-GN and Sign-IDD neglect fine-grained dependencies, their generated poses are less accurate and natural, especially in the zoomed-in regions (a) and (b), where densely connected hand joints are difficult to model precisely. In contrast, FGDM dynamically captures joint-level dependencies through the Focal stage, producing more natural and GT-consistent hand shapes and overall poses. Notably, in panels (b) and (c), even when the GT is corrupted by occlusion or motion blur, FGDM still produces results that are closer to the raw image. This robustness partly arise from our SCG strategy, which semantically mitigates the impact of inaccurate GT during training. Overall, FGDM demonstrates superior realism and robustness with previous methods.

Qualitative comparison between w/o and w/ Focal. As shown in Fig. 5, without the Focal stage, the model struggles to capture joint-level dependencies, resulting in unnatural and distorted hand shapes due to the complex correlations among hand joints. In contrast, with the Focal stage, the model learns intricate joint relationships through the ASGCN and generates more natural and accurate gestures—for example, correctly producing the extended index finger in the first case and the clenched fist in the second. These results demonstrate that the Focal stage effectively models joint-level dependencies and enhances fine-grained pose accuracy.

5. Conclusions

This paper presents a novel Focal–General Diffusion Model (FGDM) with a Semantic Consistent Guidance (SCG) mechanism for Sign Language Production (SLP). Specifically, FGDM introduces a pioneering two-stage denoising framework that effectively balances joint-level dependencies and global coherence. In the Focal stage, we propose the Adaptive Sign GCN (ASGCN), which reformulates graph convolution to enable frame-wise adaptive pose modeling. Furthermore, SCG seamlessly integrates semantic supervision into diffusion training, enforcing tighter semantic consistency between generated poses and gloss sequences. Extensive quantitative and qualitative experiments demonstrate that FGDM surpasses existing SOTA methods by a clear margin. We hope this work can inspire future research to further explore the potential of diffusion frameworks in SLP from new perspectives.

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